

Chapter 3: Angles as Turns

Worksheet 3: Angles as Turns

Learning Objective: Recognise angles as turns/rotations; identify right, acute, obtuse and straight angles; relate angles to clock hands and directions.

Worked Example:

A clock's minute hand moves from 12 to 3. What type of angle (turn) does it make?

Solution: This is a quarter turn, which equals a right angle (90°).

Questions (1–20)

1. Name the angle formed when a hand makes a half turn.
2. How many degrees are there in a right angle?
3. At 6 o'clock, is the angle between the clock hands acute, right, obtuse or straight?
4. A full turn equals how many degrees?
5. Name one everyday object that has a right-angle corner.
6. Look at the corner of your notebook. What type of angle is it?

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7. A door is opened so that it makes a quarter turn from its closed position. What type of angle does it form with the wall?

 8. At 9 o'clock, what type of angle do the clock hands make?

 9. A wheel makes one and a half full turns. How many degrees has it turned in total?

 10. If you face North and turn a quarter turn clockwise, which direction do you face?

 11. Classify these angles as acute, right, obtuse or straight: 30° , 90° , 120° , 180° .

 12. A football player kicks the ball and it changes direction by a half turn. How many degrees is that?

 13. The hour hand of a clock moves from 12 to 4. What fraction of a full turn is this, and how many degrees?

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14. At 7 o'clock, the angle between the clock hands is approximately 150° . Is this acute, right, obtuse or straight?
15. A gymnast spins making three quarter turns. What is the total angle of rotation?
16. Door A opens to 60° and Door B opens to 120° . Which door makes an obtuse angle with the wall, and by how many degrees is Door B's angle bigger than Door A's?
17. A ceiling fan blade completes 2 full turns in one second. How many degrees does it rotate in one second?
18. The minute hand of a clock makes a full turn (360°) in 60 minutes. How many degrees does it turn in 15 minutes?
19. While playing a sport (cricket, badminton, kabaddi) or dancing, describe a moment from your day when you made a half-turn or a quarter-turn.

20. Stand facing your school gate (or main door at home). If you make a half turn, which direction will you now be facing? Try it and write what you observe.

Reflection Question: Look around your classroom and find three angles: one that looks like a right angle, one smaller, and one bigger. Write what objects they are.

Real-Life Application Question: On your way home today, count how many right-angle turns (corners) you take. Write the number and briefly describe the path.